MTH2003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Differential Equations

Module Leader: Prof. Vadim N. Biktashev

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count

USEFUL FORMULAS

These can be used without proof if and where needed.

Classical Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right], \quad x \in [A, A+2L],$$

$$a_0 = \frac{1}{2L} \int_A^{A+2L} f(x)\,\mathrm{d}x, \quad a_n = \frac{1}{L} \int_A^{A+2L} f(x) \cos\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x,$$

$$b_n = \frac{1}{L} \int_A^{A+2L} f(x) \sin\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x.$$

Fourier sine-series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right), \quad x \in [0, L],$$

$$b_n = \frac{2}{L} \int_0^{L} f(x) \sin\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x.$$

Fourier cosine-series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right), \quad x \in [0, L],$$

$$a_0 = \frac{1}{L} \int_0^{L} f(x)\,\mathrm{d}x \quad a_n = \frac{2}{L} \int_0^{L} f(x) \cos\left(\frac{n\pi x}{L}\right)\,\mathrm{d}x.$$

Useful integrals

$$\int \frac{\sin(x)}{\cos(x)}\,\mathrm{d}x = -\ln|\cos(x)| + C, \qquad \int \frac{\cos(x)}{\sin(x)}\,\mathrm{d}x = \ln|\sin(x)| + C,$$

$$\int \frac{\sin^3 x}{\cos^2 x}\,\mathrm{d}x = \cos(x) + \frac{1}{\cos(x)} + C,$$

$$\int \frac{\mathrm{d}x}{\sin^2(x)\cos(x)} = -2\frac{1}{\sin(x)} + \ln\left|\frac{1+\sin(x)}{1-\sin(x)}\right| + C.$$

SECTION A

1. (a) Find and plot the solutions of

$$(1+\cos x)\frac{\mathrm{d}y}{\mathrm{d}x} + y\sin x = 0$$

subject to initial conditions: (A) $y(0) = 0$, (B) $y(0) = \pi$, (C) $y(\pi) = 0$, (D) $y(\pi) = \pi$. For each solution, identify the maximal interval for which it is unique and show that in the plot where appropriate. For the plots, use ranges $x \in [-2\pi, 2\pi]$, $y \in [-2\pi, 2\pi]$. (15)

(b) Verify that $y_1(x) = \sin(x)$ is a solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \tan(x)\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$$

and use that to find the general solution of this equation. Find also the solution satisfying $y(0) = 0$, $y'(0) = 1$, and state the maximal interval in which this solution is defined. (15)

(c) Consider the eigenfunction problem

$$x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} + \lambda x^3 y = 0,\ x \in [0,1]; \qquad y(0) = 0,\ y(1) = 0.$$

Bring this equation into the Sturm-Liouville form and hence state the orthogonality condition for the eigenfunctions of this problem. Verify that $y_* = \sin(4\pi x^2)$ is an eigenfunction of this problem and find the corresponding eigenvalue λ_*. State how many eigenvalues smaller than λ_* this problem has. *This problem is singular at $x = 0$; assume without proof that any results of Sturm-Liouville theory you may need are applicable here notwithstanding.* (10)

(d) Use a trial function of the form $y = \sum_{n=0}^{\infty} a_n x^n$ to find the solution of the differential equation

$$(10 + x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + y = 0.$$

Write the general solution in the form $y = Ay_1(x) + By_2(x)$, where $y_1(0) = 1$, $y_1'(0) = 0$ and $y_2(0) = 0$, $y_2'(0) = 1$. Write down the first three non-zero terms of the power series for $y_1(x)$ and $y_2(x)$. Determine the radius of convergence of the resulting power series. (10)

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SECTION B

2. Consider the system of equations

$$\text{(Eq2)} \qquad \begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= y + g_1(t), \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= -x + g_2(t), \end{aligned}$$

where $g_1(t)$, $g_2(t)$ are given functions of time.

(a) Write system (Eq2) in the vector form $\mathrm{d}\mathbf{x}/\mathrm{d}t = \mathbf{A}\mathbf{x} + \mathbf{g}$. Identify $\mathbf{x}$, $\mathbf{A}$ and $\mathbf{g}$. Find eigenvalues and eigenvectors of the matrix of coefficients $\mathbf{A}$. Hence write down the general solution $x(t)$, $y(t)$ of (Eq2) for $g_1(t) \equiv 0$, $g_2(t) \equiv 0$. Make sure to present the general solution in the real (i.e. not complex) form. Write down the fundamental matrix $\mathbf{\Psi}(t)$ of this system, satisfying the initial condition $\mathbf{\Psi}(0) = \mathbf{I}$. (10)

(b) Using the method of variation of parameters, find the solution to (Eq2) for $g_1 = \tan^2 t - 1$, $g_2 = \tan t$, for initial condition $x(0) = 0$, $y(0) = 2$. Identify the maximal interval of t on which the solution exists. (10)

(c) Draw the phase portrait of the homogeneous system based on your results of part (a). Classify this phase portrait (stable node, unstable node, saddle, stable focus, unstable focus or center). On the same plot, draw the curve corresponding to the solution to the non-homogeneous system obtained in part (b). (5)

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3. Consider the following ODE, known as the spherical Bessel equation of order 0:

(Eq3) $$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{2}{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0.$$

(a) Explain why $x = 0$ is a regular singular point of this equation. Find all values of m for which solutions of this ODE in the form

$$y(x) = x^m \sum_{n=0}^{\infty} a_n x^n, \qquad a_n = \text{const}, \qquad a_0 \neq 0$$

exist. (6)

(b) Find the transformation of the form

$$y(x) = q(x)z(x)$$

where $z(x)$ is the new unknown function, and $q(x) \neq 0$ is a fixed function you need to define, such that the ODE for $z(x)$ does not have the term with the first derivative $\mathrm{d}z/\mathrm{d}x$. Solve the resulting ODE for $z(x)$ and hence find the general solution of (Eq3). Show that this general solution is in agreement with the result of part (a) about allowed values of m. (12)

(c) Using the results of part (b), show that all solutions of the eigenvalue problem on $x \in (0, L]$,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{2}{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \lambda y = 0, \quad |y(0)| < \infty, \quad y(L) = 0,$$

for $\lambda > 0$ are given by

$$\lambda_n = \frac{n^2\pi^2}{L^2}, \qquad y_n(x) = \frac{1}{x}\sin\left(\frac{n\pi x}{L}\right), \qquad n = 1, 2, 3\ldots$$

(7)

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4. Cooling of a spherical ball of radius R is described by the heat conduction equation, which in spherical coordinates takes the form

$$\frac{\partial u}{\partial t} = \frac{1}{r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{\sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial u}{\partial \phi}\right) + \frac{1}{\sin^2\phi}\frac{\partial^2 u}{\partial \theta^2}\right],$$

where $u(r, \theta, \phi, t)$, defined for $r \leq R$ and $t \geq 0$, is the temperature at the time moment t at the point at the distance r from the centre in the direction determined by the spherical angles θ, ϕ. The temperature distribution at all times is spherically symmetric, that is, $u(r, \theta, \phi, t) = v(r, t)$. At the surface of the ball, $r = R$, the temperature is maintained at the constant level $u = 0$.

(a) Write down the boundary-value problem for the function $v(r, t)$, corresponding to the physical situation described above. *Note that the formula for the Laplacian in spherical coordinates given above includes division by r so the limit $r \to 0$ requires special attention.* (5)

(b) Using separation of variables, find the general solution to the boundary-value problem you formulated in part (a) as an infinite series of the form

$$v(r, t) = \sum_{n=1}^{\infty} X_n(r) W_n(t),$$

where $X_n(r)$ and $W_n(t)$ are functions you need to identify. Assume without proof that all solutions satisfy $v(r, t) \to 0$ as $t \to \infty$. *You may use, without re-deriving, the result of Question 3c above, even if you have not obtained it.* (11)

(c) Find also the particular solution to the above problem, corresponding to the initial condition

$$v(r, 0) = v_0(r), \qquad r \in (0, R],$$

where $v_0(r)$ is a given function. Assume that $v_0(r) > 0$ for all $r \in (0, R)$ and show that in the limit $t \to \infty$, asymptotically

$$v(r, t) \approx X(r)\, \mathrm{e}^{-\mu t},$$

where you are requested to find $X(r)$ and μ. (9)

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